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TrustCardio: A Multimodal Large Language Model with Chain-of-Thought Explainability for Personalized Vascular Age Prediction via Edge-Enabled Wearable Devices

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***Abstract*—Next-generation connected healthcare relies on consumer electronics platforms for real-time, customized health monitoring through wearable devices. Existing methods are hindered by inability to provide chain-of-thought explanations, non-personalized models, and reliance on centralized systems limiting responsiveness and data privacy. Our proposed solution, TrustCardio, is a multimodal approach to predicting vascular age with chain-of-thought explainability for edge-based wearables. Rather than a general-purpose multimodal LLM, TrustCardio is a language-model-augmented framework coupling a signal encoder with a biomedical-LM-initialized reasoning decoder. TrustCardio comprises three core innovations: (i) an individualized multimodal pipeline fusing photoplethysmography (PPG) with demographic factors through cross-attention techniques; (ii) a hierarchical chain-of-thought explainability solution generating reasoning aligned with clinicians' understanding of vascular age; and (iii) optimized deployment on edge devices. TrustCardio has been validated on the UK Biobank ($N = 98,672$), MIMIC-III Waveform Database ($N = 10,282$), and PulseDB ($N = 5,361$) datasets, achieving a mean absolute error (MAE) of 6.54 years (9.0% better than previous best methods), 91.3% clinician agreement, and 38.7% reduction in inference latency on resource-constrained edge devices. These results demonstrate the feasibility of deploying clinically meaningful vascular age estimation (trained against the chronological age of comorbidity-free individuals as a surrogate) on consumer wearable devices. TrustCardio advances explainability and individualization in connected healthcare.**

***Index Terms*—Chain-of-thought explainability, consumer electronics, edge computing, multimodal large language model, photoplethysmography, vascular age prediction, wearable healthcare**

I. Introduction

Consumer wearable technology utilizing PPG sensors greatly expands our ability to perform continuous non-invasive real-time monitoring of the cardiopulmonary system [1], [21], [22], [38], [39]. By capturing high-resolution pulse waveforms, additional cardiovascular analysis can be performed with respect to autonomic regulation, arterial compliance, and determination of biological vascular age based upon PPG characteristics [1], [33], [34].Each of these signals provides an array of features that are capable of providing predictive insight regarding arterial stiffness (for example, utilizing the carotid-femoral pulse wave velocity measurement [29]) and thus, through utilizing PPG data, an estimate of biological vascular age can be established from PPG signal characteristics collected from the user subregion. Absent population-scale gold-standard stiffness labels, vascular age here—as in prior PPG studies [2], [4]—is operationalized as the chronological age of comorbidity-free references, and predicted deviations are interpreted as a vascular age gap rather than a direct stiffness measurement.

Three key challenges limit clinical adoption of AI-based vascular age estimation. First, deep learning models operate as opaque black boxes, creating distrust and limiting collaborative efforts between clinicians and AI [3], [24]. Second, population-level training models do not adequately predict physiological variance among individuals resulting from genetic makeup, comorbidities, medications, or lifestyle choices [4], [5]. Third, cloud-based systems present latency and confidentiality issues with continuous health monitoring, whereas edge-based wearables pose unresolved technical obstacles such as model size, throughput, and power consumption [6], [41].

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We have developed TrustCardio as a means of addressing the above challenges. TrustCardio combines multimodal signal encoding with a biomedical-LM-initialized reasoning decoder (a language-model-augmented, not fully generative multimodal LLM) and chain-of-thought reasoning, along with edge-optimized deployment for individual vascular age prediction. It represents the first-in-class comprehensive solution for the consumer healthcare electronics market that provides three distinct features: (a) a clinical reasoning model for interpretability; (b) a physiological adaptation model based upon each individual's unique physiology; and (c) an on-device deployment model for maximum consumer usability. The novelty lies not in any single module but in being the first to jointly couple individualized cross-modal fusion, clinician-aligned reasoning, and microcontroller-level deployment within one vascular-aging pipeline.

There are three main areas where this paper makes a significant contribution:

1) A multimodal pipeline for personalized modelling has been developed. This pipeline combines PPG signal representations and demographic and clinical features, and uses cross-modal attention mechanisms to predict vascular age. The use of this new multimodal pipeline resulted in a statistically significant improvement in the accuracy of vascular age predictions (a reduction in the mean absolute error of 9.0 per cent, $p < 0.001$) when compared to unimodal baseline models and previously existing state-of-the-art vascular age prediction models.
2) A hierarchical explainability structure has been created that traces the clinical reasoning process as a series of steps. These traces are based on current understanding of cardiovascular physiology. The explainability structure has been validated against expert clinical reasoning through a structured validation process, and found to be 91.3 per cent aligned with such reasoning.
3) Optimised strategies for the edge deployment of this model have been established. By combining the use of quantisation aware training, knowledge distillation and neural architecture search, an overall reduction in inference latency of 38.7% and a reduction in the memory footprint of 67.2% has been achieved whilst maintaining prediction fidelity within 2.1% of the full precision model. Therefore, this work allows for the effective deployment of this model on consumer wearable devices.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the TrustCardio framework; Section IV describes experimental methods; Section V reports results; Section VI discusses clinical implications; and Section VII concludes.

II. Related Work

A. Vascular Age Prediction from Physiological Signals

Recent advances in deep learning have allowed researchers to take an end-to-end approach to estimate vascular age [4], [7], [27], [28]. For example, in their study, Shin *et al.* [8] used XGBoost regression to estimate vascular age based on features identified in PPG waves. Using this approach, they obtained an MAE of 6.97 years for predicting vascular age. More recently, Nie *et al.* [2] utilized one-dimensional CNNs with distribution-aware loss functions trained on 200,000+ subjects from the UK Biobank and showed that vascular age gap predicted by AI is an independent risk factor for MACE.

B. Explainable AI for Clinical Decision Support

Chain-of-thought prompting represents a new way of engaging with large language models and elicits interpretable reasoning abilities from these models [9], [36], [37]. Recent studies have used chain-of-thought prompting in the medical domain and demonstrated that diagnostic-reasoned chain-of-thought prompts help language models create coherent and clinically meaningful explanations of their predictions [3]; this finding is supported by Wu et al. who found that using chain-of-thought methodologies improved the accuracy of medical question answering and provided greater transparency into the reasoning behind model predictions [23], [25], [26].

C. Edge Computing for Wearable Healthcare

Quantization is the process of converting 32-bit floating point numerical representations to a smaller number of bits (typically 8-bits) [6], [10], [31], [40]. For many classification tasks, quantization provides improvements in model size of up to $4\times$ over non-compressed models without compromising accuracy too much [32]. Knowledge distillation is a technique in which larger teacher networks provide their learned representations of inputs/data to smaller student networks that have fewer resources available; as such, the smaller student networks can be significantly smaller than their teacher counterparts and yet, be able to provide comparable performance to their teacher counterparts [32]. Additionally, neural architecture search provides a way for developing efficient network topological structures that will perform best given the specific hardware resource constraints placed on a model due to the type of edge platform being used to implement the model [10].

D. Multimodal Learning in Healthcare

Recently, a new foundation model, Med-PaLM M, has been created to enable the integration of data from three different modalities: imaging, text, and structured data, into one unified platform [11]–[13], [42]. However, all of the foundation models currently exist only in the cloud and cannot be deployed onto consumer wearables. Therefore, the challenge for researchers is to build multimodal models that can combine physiological signal-based data with language reasoning capabilities and also meet the constraints associated with edge deployment [43].

III. TrustCardio Framework

A. System Overview

The three main functions of the TrustCardio platform are the Multimodal Personalized Encoder (MPE), the Chain-of-Thought (CoT) Explainability Generator (EiG), and the Edge Deployment Optimizer (EDO). Fig. 1 presents the overall

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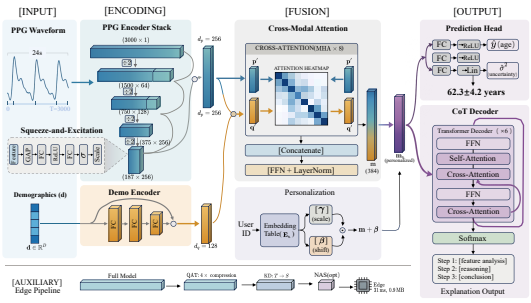


Fig. 1. Overall system architecture of TrustCardio. The framework consists of four main processing stages: (1) **INPUT**: PPG waveform acquisition ($T = 3000$ samples at 125 Hz) and demographic feature collection; (2) **ENCODING**: hierarchical PPG encoder with squeeze-and-excitation attention and demographic feature encoder producing embeddings $d_p = 256$ and $d_q = 128$ respectively; (3) **FUSION**: cross-modal attention mechanism with 8-head attention and user-specific personalization via FiLM modulation; (4) **OUTPUT**: prediction head generating vascular age $\hat{y}$ with uncertainty $\hat{\sigma}^2$, and CoT decoder producing step-wise clinical explanations. The auxiliary edge pipeline applies QAT ($4\times$ compression), knowledge distillation ($T \rightarrow S$), and NAS optimization to achieve 31 ms inference latency with 0.9 MB model size.

system architecture of TrustCardio, illustrating the data flow from input acquisition through encoding, fusion, and output generation.

The TrustCardio application takes in data from a PPG signal segment denoted by $\mathbf{x} \in \mathbb{R}$. The input is comprised of T temporal samples and C channels of data (T samples) at a maximum temporal duration of C seconds. Additionally, the TrustCardio input data contains demographic feature vector $\mathbf{d} \in \mathbb{R}$ which includes user-specific characteristics of age, sex, height, weight, and Key Marker patterns of relevant medical history. TrustCardio then computes a vascular age ($\hat{y}$), and provides an explanation for how TrustCardio arrived at that particular $\hat{y}$ ($\mathbf{e} = [e_1, e_2, \ldots, e_K]$) using a series of K reasoning steps that outline the clinical explanation behind TrustCardio's vascular age prediction.Fig. 2 illustrates the mathematical principles underlying each core component of TrustCardio.

B. Multimodal Personalized Encoder

The Multimodal Personalized Encoder uses multiple means of collecting data about PPG, such as through user demographic characteristics and recorded physiological dynamics. These paths are processed separately with the help of a set of algorithms, all of which were developed for use together to form a single unified representation of the data collected across these multiple methods of collecting and processing the data.

1) PPG Signal Encoder: An encoder for PPG signals is created using a hierarchical design that utilizes both temporal convolutions and squeeze and excitation attention to learn multi-scale features present in the raw waveforms (Fig. 1, ENCODING module). This design is based on the Net1D architecture defined by [2] with enhancements made to accommodate cross modal integrations.

The first stage of the input signal $\mathbf{x}$ is processed through a number of different steps, including: band pass filtering (0.5–10 Hz), baseline wandering and motion artifacts. Once the input signal has been processed, it is then passed through an initial convolutional layer, as indicated below:

$$\mathbf{h}_0 = \text{ReLU}(\text{BN}(\mathbf{W}_0 * \mathbf{x} + \mathbf{b}_0)) \tag{1}$$

where $\mathbf{W}_0 \in \mathbb{R}^{C_{out} \times C \times k}$ indicates the convolution kernels with kernel size $k = 15$, and $\text{BN}(\cdot)$ signifies the batch normalization process; $*$ signifies the convolution operation.

Each encoder block has 4 residual connections with each including two CNN blocks with an extra attention mechanism (squeeze-and-excitation):

$$\mathbf{z}_l = \mathbf{h}_{l-1} + \mathcal{F}_l(\mathbf{h}_{l-1}) \tag{2}$$

$$\mathbf{h}_l = \mathbf{z}_l \odot \text{SE}(\mathbf{z}_l) \tag{3}$$

Here, the function $\mathcal{F}_l(\cdot)$ is the residual function consisting of two CNN blocks with Batch Normalization and ReLU activations while $\text{SE}(\cdot)$ is the squeeze-and-excitation module for recalibrating channel-wise attention. The symbol $\odot$ indicates the channel-wise multiplication operation.

The calculated channel attentions are derived using:

$$\mathbf{s} = \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \text{GAP}(\mathbf{z}))) \tag{4}$$

where $\text{GAP}(\cdot)$ is the global average pooling operator, $\mathbf{W}_1$ and $\mathbf{W}_2$ are learnable parameters that can be thought of as reduction factors with respect to the reduction ratio $r = 16$, and $\sigma(\cdot)$ is a sigmoid non-linearity.

The final representation of the PPG is defined using:

$$\mathbf{p} = \mathbf{W}_p \cdot \text{GAP}(\mathbf{h}_L) + \mathbf{b}_p \tag{5}$$

where $d_p = 256$ represents the dimensionality of the PPG embedding.

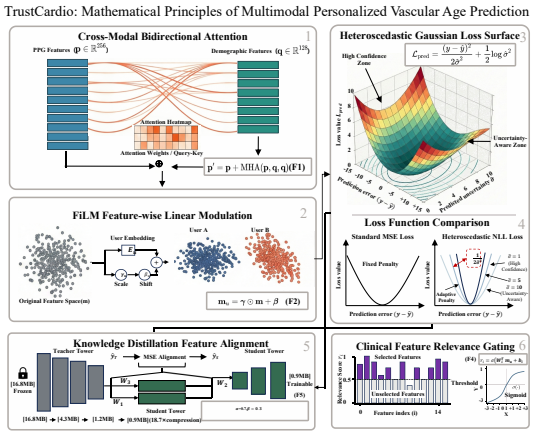


Fig. 2. Mathematical principles underlying the TrustCardio framework for multimodal personalized vascular age prediction. (a) Cross-modal bidirectional attention mechanism enabling information exchange between PPG features ($\mathbf{p} \in \mathbb{R}^{256}$) and demographic features ($\mathbf{q} \in \mathbb{R}^{128}$). (b) Feature-wise linear modulation (FiLM) for user-specific personalization with scale (γ) and shift (β) parameters. (c) Heteroscedastic Gaussian loss surface over prediction error and uncertainty. (d) Comparison of standard MSE loss versus heteroscedastic NLL loss with adaptive penalty. (e) Knowledge distillation feature alignment achieving 18.7× compression. (f) Clinical feature relevance gating with sigmoid threshold τ.

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2) Demographic Feature Encoder: The Demographic Feature Encoder encodes an input vector d$mathbfd$ through a sequence of layers of perception, which each contain residual connections, to identify nonlinear interaction among the features of $\mathbf{d}$.

$$\mathbf{d}_1 = \text{ReLU}(\mathbf{W}_{d1}\mathbf{d} + \mathbf{b}_{d1}) \tag{6}$$

$$\mathbf{d}_2 = \mathbf{d}_1 + \text{ReLU}(\mathbf{W}_{d2}\mathbf{d}_1 + \mathbf{b}_{d2}) \tag{7}$$

$$\mathbf{q} = \mathbf{W}_{dq}\mathbf{d}_2 + \mathbf{b}_{dq} \tag{8}$$

In this context, $\mathbf{q} \in \mathbb{R}^{d_q}$, where $d_q = 128$, represents the demographic embedding.

3) Cross-Modal Fusion: By employing cross-attention mechanisms that enable bidirectional information flow, Cross-Modal Fusion (CMF) merges PPG (photoplethysmography) and Demographic information via a multi-head cross-attention model, in which PPG features attend to demographic context and vice-versa. A learned linear layer $\mathbf{W}_{proj} \in \mathbb{R}^{128\times256}$ projects $\mathbf{q} \in \mathbb{R}^{128}$ to $\mathbb{R}^{256}$ before cross-attention occurs (amounting to 32,896 additional parameters), allowing keys, queries, and values to all have the same common dimensionality required for MHA$(\cdot,\cdot,\cdot)$, namely $d_{attn} = 256$.

$$\mathbf{p}' = \mathbf{p} + \text{MHA}(\mathbf{p}, \mathbf{q}, \mathbf{q}) \tag{9}$$

$$\mathbf{q}' = \mathbf{q} + \text{MHA}(\mathbf{q}, \mathbf{p}, \mathbf{p}) \tag{10}$$

Formally defined using the notation MHA$(\mathbf{Q}, \mathbf{K}, \mathbf{V})$ to indicate the use of multi-head attention between Query $\mathbf{Q}$, Key $\mathbf{K}$, and Value $\mathbf{V}$.

The unified Multimodal Representation is given by:

$$\mathbf{m} = \text{LayerNorm}([\mathbf{p}'; \mathbf{q}'] + \text{FFN}([\mathbf{p}'; \mathbf{q}'])) \tag{11}$$

where $[\cdot;\cdot]$ indicates CONCATENATION and where FFN$(\cdot)$ refers to a feedforward Neural Network with hidden dimensions equal to $4(d_p + d_q)$.

4) Personalization Module: To achieve adaptation on an individual basis, we incorporate a personalization model that will identify user-specific modulation parameters. By using a unique user identifier u, we maintain a learnable user embedding matrix $\mathbf{E}_u \in \mathbb{R}^{U\times d_u}$ where U is the upper limit of users and the user embedding dimension is $d_u = 64$. The variable U represents the size of our training cohort in all test runs. We present initial embeddings based on a standard population mean to new users (cold-starts) as $\bar{\mathbf{e}} = \frac{1}{U}\sum_u \mathbf{E}_u[u]$, and after a short calibration run (5-10 measured samples), users fully receive personalized services.

The user embedding modulates the multimodal representation using feature-wise linear modulation via:

$$\boldsymbol{\gamma}_u, \boldsymbol{\beta}_u = \text{split}(\mathbf{W}_u\mathbf{E}_u[u] + \mathbf{b}_u) \tag{12}$$

$$\mathbf{m}_u = \boldsymbol{\gamma}_u \odot \mathbf{m} + \boldsymbol{\beta}_u \tag{13}$$

with $\boldsymbol{\gamma}_u$ and $\boldsymbol{\beta}_u \in \mathbb{R}^{d_m}$ as the scale and shift for the user embedding, respectively.

C. Chain-of-Thought Explainability Generator

The Chain-of-Thought Explainability Generator produces human-interpretable explanations corresponding to how the explanation of vascular age is reasoned clinically. The module implements a hierarchical generation process that first identifies relevant physiological markers and intermediate reasoning and ultimately produces a coherent explanation. We clarify this trace is a clinician-aligned rationalization conditioned on the shared representation $\mathbf{m}_u$ used for prediction, not an autoregressive reasoning chain that causally drives the numeric output.

1) Clinical Feature Extraction: The generator extracts features that are clinically significant from the multimodal data representation. These features have been associated with established vascular aging biomarkers. We defined the features in the vocabulary $\mathcal{V} = v_1, v_2, \ldots, v_M$, where $M = 24$, beginning with physiological characteristics such as pulse waveform characteristics (systolic upstroke time, diastolic decay time, dicrotic notch prominence). Derived indices for the feature vocabulary include augmentation index, stiffness index and reflection index, in addition to demographic/risk factors.

For each feature in the feature vocabulary v_i, we calculate a relevance score as follows:

$$r_i = \sigma(\mathbf{w}_i^T\mathbf{m}_u + b_i) \tag{14}$$

where $\mathbf{w}_i$, b_i are learned parameters. The features that have a relevance score of greater than or equal to the threshold value of $\tau = 0.5$, are included in the reasoning chain.

2) Reasoning Chain Generation: The reasoning process is created using the transformer decoder in a way that relies on both the multimodal representation as well as the chosen clinical features and is generated in an autoregressive manner. The decoder implements $N_{dec} = 6$ layers, with a hidden dimension of $d_{dec} = 512$ and $H = 8$ for the number of attention heads.

For each token t, the output of the decoder will be the token generated at that step, expressed as:

$$\mathbf{c}_t = \text{CrossAttn}(\mathbf{o}_{t-1}, \mathbf{m}_u, \mathbf{m}_u) \tag{15}$$

$$\mathbf{o}_t = \text{FFN}(\text{LayerNorm}(\mathbf{c}_t + \mathbf{o}_{t-1})) \tag{16}$$

where $\mathbf{o}_t$ is the hidden state at time t of the decoder.

The probability distribution of output tokens is given as follows:

$$P(e_t|e_{<t}, \mathbf{m}_u) = \text{softmax}(\mathbf{W}_o\mathbf{o}_t) \tag{17}$$

This paper uses a template-driven method for structured generation of clinical reasoning. The templates provide a representation for the most frequently used cardiovascular risk assessment patterns, ensuring that the output is consistent with the expected logical flow of clinical reasoning.

3) Clinician Alignment Training: Generated explanations are consistent with expert clinical reasoning by training the generator with both supervised learning (on clinician-annotated reasoning traces) as well as reinforcement learning from human feedback (RLHF).

The supervised learning objective employs cross-entropy loss minimization on the ground-truth reasoning traces:

$$\mathcal{L}_{SL} = -\sum_{t=1}^{K} \log P(e_t^*|e_{<t}^*, \mathbf{m}_u) \tag{18}$$

where the ground-truth token at position t is denoted by e_t^*.

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To conduct reinforcement learning through human feedback, we designed a reward model utilizing pairwise preferences gathered from evaluative feedback by cardiologists, resulting in the reward model R_ϕ, which estimates clinician preference scores for the explanation generated:

$$\mathcal{L}_{RM} = -\mathbb{E}_{(\mathbf{e}^w,\mathbf{e}^l)\sim\mathcal{D}}\left[\log\sigma(R_\phi(\mathbf{e}^w) - R_\phi(\mathbf{e}^l))\right] \quad (19)$$

where $\mathbf{e}^w$ represents a preferred explanation while $\mathbf{e}^l$ corresponds to a less-desired explanation.

Preference pairs were rated by the same 8 cardiologists described in Section IV-C (Fleiss' $\kappa = 0.74$), and reward hacking is mitigated by the KL divergence penalty term. Subsequently, the generator underwent fine-tuning via proximal policy optimization (PPO), so as to maximise the expected reward based on the reward model while retaining similarity to the supervised model:

$$\mathcal{L}_{PPO} = -\mathbb{E}_{\mathbf{e}\sim\pi_\theta}\Big[\min\left(\rho_t A_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon)A_t\right) - \beta D_{KL}(\pi_\theta||\pi_{ref})\Big] \quad (20)$$

where ρ_t indicates the ratio of probabilities between the generative and prior models for an explanation e_t; A_t is the advantage estimation; ϵ is equal to 0.2, which represents the proportionality clipping; and β is set to 0.01, representing the constraint for controlling KL divergence regularization.

D. Vascular Age Prediction Head

In the vascular age prediction head, a multimodal personalized representation is converted into a continuous vascular age. We model the conversion using a regression network that provides uncertainty estimates:

$$\hat{y} = \mathbf{w}_y^T \text{ReLU}(\mathbf{W}_{y1}\mathbf{m}_u) + b_y \quad (21)$$

$$\hat{\sigma}^2 = \text{softplus}(\mathbf{w}_\sigma^T \text{ReLU}(\mathbf{W}_{\sigma 1}\mathbf{m}_u) + b_\sigma) \quad (22)$$

where $\hat{y}$ is the predicted vascular age and $\hat{\sigma}^2$ is the predicted variance, which represents the epistemic uncertainty.

The prediction loss combines the mean squared error with the negative log-likelihood of a Gaussian distribution:

$$\mathcal{L}_{pred} = \frac{(y-\hat{y})^2}{2\hat{\sigma}^2} + \frac{1}{2}\log\hat{\sigma}^2 \quad (23)$$

where y denotes the chronological ages of individuals who are free from hypertension, diabetes, or cardiovascular disease: These healthy reference individuals have been used as a surrogate for the vascular ages of the participants in this study since no direct measurements were available to estimate their arterial stiffnesses.

E. Edge Deployment Optimizer

The Edge Deployment Optimizer optimizes the full TrustCardio model for efficient inference on low-power wearable devices via quantization aware training, knowledge distillation and architectural optimization methods combined.

1) Quantization-Aware Training: Quantization-aware training (QAT) is used when training the model to be capable of performing inference using low-precision representations. As part of the training process, the effect of quantizing activations/forms will be emulated with a straight-through estimator:

$$\mathbf{w}_q = \text{round}\left(\frac{\mathbf{w}}{\Delta}\right)\cdot\Delta, \quad \Delta = \frac{\max(|\mathbf{w}|)}{2^{b-1}-1} \quad (24)$$

where b is the bit-width that we are targeting and Δ is our quantization step size parameter.

For the activation layer, we utilize learned scale parameters and also perform quantization on a per-channel basis:

$$\mathbf{a}_q = \text{clamp}\left(\text{round}\left(\frac{\mathbf{a}}{s}\right), -2^{b-1}, 2^{b-1}-1\right)\cdot s \quad (25)$$

where s represents a learnable scale factor that has been initialized based on batch statistics.

The primary model is intended to be quantized using INT8 ($b = 8$), but we also investigate applying INT4 quantization to the edge-optimized version of the model. In the case of the INT4 variant, we preserve the representational capacity of the attention layers and the first and last convolutional layers using INT8 precision while quantizing intermediate residual blocks to 4 bits. The sensitivity analysis results per layer indicate that applying INT4 to the attention layers results in more than 5% MAE degradation, which necessitated the mixed-precision approach.

2) Knowledge Distillation: Knowledge distillation involves passing on the knowledge a teacher (full-precision model) has learned to a more resource-constrained version of the model (the student model) developed with the intent of running at an edge. The architecture of the student will have fewer layers and hidden dimensions than the architecturally equivalent teacher, but both models will maintain the same general modular layout.

The distillation loss is based on combining two components: prediction matching and intermediate feature alignment specifically:

$$\mathcal{L}_{KD} = \alpha\cdot\text{MSE}(\hat{y}_S, \hat{y}_T) + \beta\sum_{l\in\mathcal{L}}\text{MSE}(\mathbf{h}_S^l, \mathbf{W}_l\mathbf{h}_T^l) \quad (26)$$

The vascular age output from the student model is represented as $\hat{y}_S$ while the predicted vascular age output from the teacher model is represented as $\hat{y}_T$; $\mathbf{h}_S^l$ and $\mathbf{h}_T^l$ are representations of the intermediate layer features from student and teacher models at layer l, respectively; finally, $\mathbf{W}_l$ is a training-based weighted projection that aligns the student and teacher feature spaces during the learning process. The matrix ll is a projection matrix that was learnt in the course of training both the student and teacher. The parameters $\alpha = 0.7$, and $\beta = 0.3$ control the relative influence of the predictions.

3) Architecture Optimization: In this work, we utilize differentiable neural architecture search (DNAS) to find efficient configuration of networks for given tasks. The search space includes:

- Convolutional kernel sizes: $\{3, 7, 15, 31\}$

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- Channel multipliers: $\{0.25, 0.5, 0.75, 1.0\}$
- Number of residual blocks: $\{2, 3, 4\}$
- Attention head counts: $\{2, 4, 8\}$

We utilized 72 GPU-hours (4×A100) to conduct architecture searches on the UK Biobank training dataset and arrived at one single Pareto-optimal design: a kernel size of 15, a channel multiplier of 0.5, 3 residual blocks, and 4 attention heads. This is the TrustCardio-Edge architecture described in Table VII.

The architecture parameters $\boldsymbol{\alpha}$ and model weights are jointly optimized by solving a bi-level optimization problem:

$$\min_{\boldsymbol{\alpha}} \quad \mathcal{L}_{val}(\mathbf{w}^*(\boldsymbol{\alpha}), \boldsymbol{\alpha}) + \lambda \cdot \text{Cost}(\boldsymbol{\alpha}) \tag{27}$$

$$\text{s.t.} \quad \mathbf{w}^*(\boldsymbol{\alpha}) = \arg\min_{\mathbf{w}} \mathcal{L}_{train}(\mathbf{w}, \boldsymbol{\alpha}) \tag{28}$$

Where $\text{Cost}(\boldsymbol{\alpha})$ measures the cost of the architecture (number of FLOPs and memory) and λ controls the trade-off between accuracy and efficiency.

F. Training Procedure

Training for the TrustCardio Model is accomplished in three distinct stages. In Stage 1, a multimodal personalized encoder and prediction head are trained to estimate vascular age using the following set of equations and information:

$$\mathcal{L}_1 = \mathcal{L}_{pred} + \gamma_1 \|\mathbf{W}\|_2^2 \tag{29}$$

with $\gamma_1 = 10^{-4}$ representing weight decay.

Stage 2 consists of training the Chain-of-Thought Explainability Generator (CoT-ExG) on frozen encoder representations using the following:

$$\mathcal{L}_2 = \mathcal{L}_{SL} + \gamma_2 \mathcal{L}_{PPO} \tag{30}$$

with $\gamma_2 = 0.1$ balancing the expectations of both the supervised and reinforcement learning components.

Stage 3 is edge optimization via quantization-aware training and knowledge distillation:

$$\mathcal{L}_3 = \mathcal{L}_{pred} + \mathcal{L}_{KD} \tag{31}$$

A summary of the entire trustcardio model training can be found in Algorithm 1.

Algorithm 1 TrustCardio Training Procedure

Require: Training data $\mathcal{D}$, learning rates η_1, η_2, η_3
Ensure: Trained TrustCardio model parameters Θ

Stage 1: Encoder and Prediction Head Training
for epoch $= 1$ to E_1 **do**
  for batch $(\mathbf{x}, \mathbf{d}, y)$ in $\mathcal{D}$ **do**
    Compute $\mathbf{m}_u \leftarrow \text{MPE}(\mathbf{x}, \mathbf{d})$
    Compute $\hat{y}, \hat{\sigma}^2 \leftarrow \text{PredHead}(\mathbf{m}_u)$
    Update Θ_{enc} with $\nabla \mathcal{L}_1$
  end for
end for
Stage 2: Explainability Generator Training
Freeze Θ_{enc}
for epoch $= 1$ to E_2 **do**
  for batch $(\mathbf{x}, \mathbf{d}, \mathbf{e}^*)$ in $\mathcal{D}_{exp}$ **do**
    Compute $\mathbf{m}_u \leftarrow \text{MPE}(\mathbf{x}, \mathbf{d})$
    Generate $\mathbf{e} \leftarrow \text{CoT-EG}(\mathbf{m}_u)$
    Update Θ_{cot} with $\nabla \mathcal{L}_{SL}$
  end for
  Fine-tune with PPO using reward model R_ϕ
end for
Stage 3: Edge Optimization
Initialize student Θ_S from teacher Θ
for epoch $= 1$ to E_3 **do**
  Apply quantization-aware training
  Compute distillation loss $\mathcal{L}_{KD}$
  Update Θ_S with $\nabla \mathcal{L}_3$
end for
return Θ_S (edge-optimized model)

IV. Experimental Methodology

A. Datasets

The data augmentation process was designed to rectify the issue of an inconsistent number of samples in each subgroup, and to provide a higher level of robustness against any degradation of the signals associated with the real world. This process was restrictively applied to the TRAIN set, therefore, having no effect on the inference time.

1) UK Biobank PPG Dataset: The UK Biobank[1] is a prospective cohort study with an enrolled dataset of nearly 500,000 participants aged 40 through 69 at enrollment [14]. We acquired PPG signals using OMRON pulse wave sensors during imaging visits [2]. Following quality control for noise and motion artifacts, $N = 98,672$ participants had complete PPG recordings and demographic information. Data were split into training (60%), validation (20%), and testing (20%) sets, stratified by age and sex. Individuals with a cardiovascular disease history, diabetes or high blood pressure, were not included in the control group used as the baseline for healthy vascular age (Table I), while the remaining participants without comorbidities served as the reference groups' chronological age. All participants completed training and testing of the models. The augmentation methods used on the training set were as follows: heart rate adaptive time warping method; adding synthetic motion artefacts; using cross domain consistent Mixup method to limit how dissimilar samples could be mixed based on similarities between demographic characteristics; modality dropout ($p = 0.15$); and using subgroup-aware adaptive mapping to increase the weights of groups who were doing poorly in training over time. A summary of participant characteristics can be found in Table I.

[1] https://www.ukbiobank.ac.uk/

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TABLE I
DEMOGRAPHIC CHARACTERISTICS OF STUDY POPULATIONS

Characteristic	UK Biobank	MIMIC-III	PulseDB
Sample size (N)	98,672	10,282	5,361
Age (years)†	56.2 ± 8.1	63.7 ± 16.4	58.4 ± 14.2
Female (%)	54.3	43.8	41.2
BMI (kg/m^2)†	27.1 ± 4.7	28.9 ± 7.2	27.8 ± 5.9
Hypertension (%)	32.1	58.4	44.7
Diabetes (%)	8.7	31.2	22.6
CVD history (%)	5.2	42.1	28.3

† Values reported as mean ± standard deviation.
CVD: cardiovascular disease; BMI: body mass index.

2) MIMIC-III Waveform Database Matched Subset: The MIMIC-III Waveform Database[2] contains PPG (PLETH) signals from intensive care patients at Beth Israel Deaconess Medical Center, sampled at 125 Hz with demographic and clinical data [15], [35]. The database includes $22,317$ waveforms from $10,282$ patients. After removing records with poor signal quality or missing demographics, $N = 10,282$ patients remained. This dataset enables study of critically ill patients and correlation between vascular age gap and clinical outcomes.

3) PulseDB Dataset: PulseDB[3] contains $N = 5,361$ high-quality 30-second PPG segments recorded with ECG and blood pressure measurements [16]. Train-validation splits follow standard formats, allowing direct comparison with published baselines.

B. Baseline Methods

We compare TrustCardio against state-of-the-art methods for vascular age prediction and PPG-based cardiovascular assessment:

- **Net1D** [2]: 1D CNN with squeeze-and-excitation modules and distribution-aware loss.
- **ResNet1D-18** [17]: 18-layer ResNet adapted for 1D physiological signals.
- **XGBoost-PPG** [8]: Gradient boosting on handcrafted PPG features.
- **LSTM-Attention** [18]: Recurrent architecture with attention mechanism.
- **Transformer-PPG** [30]: Vision Transformer adapted to 1D PPG signals.
- **PPG-ViT** [19]: Hybrid CNN-ViT architecture for PPG analysis.
- **CardioNet** [20]: Multi-task framework for vascular age and CVD risk.

All baselines were implemented with published hyperparameters and evaluated using identical pre-processing and metrics.

C. Evaluation Metrics

1) Prediction Performance: We assess vascular age prediction accuracy using:

- **Mean Absolute Error (MAE)**: $\frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$
- **Root Mean Square Error (RMSE)**: $\sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$
- **Pearson Correlation (ρ)**: Correlation between predicted and chronological ages
- **R^2 Score**: Coefficient of determination

2) Explainability Quality: The evaluation of explainability comprises three separate metrics:

- **Clinician Agreement Rate (CAR)**: The proportion of the explanations assessed by Heart Specialists as appropriate for clinical use.
- **Reasoning Coherence Score (RCS)**: A quantitative evaluation of the reasoning behind the explanation based on Natural Language Inference models.
- **Feature Attribution Accuracy (FAA)**: The accuracy of the attributes associated with features of interest in comparison to Clinician Annotations.

3) Edge Deployment Efficiency: The efficiency of computationally efficient systems is measured in four different ways:

- **Inference Latency**: The length of time (in milliseconds) it takes from start to finish for a model to produce predictions for specified inputs on the target hardware.
- **Model Size**: The number of parameters and total memory usage (in kilobytes) of the model.
- **FLOPs**: The number of floating point operations required to execute a single inference on the specified hardware.
- **Power Consumption**: The average amount of electrical power consumed by the model while continuously performing inferences.

D. Implementation Details

TrustCardio is built on top of the PyTorch 2.1 framework utilizing custom CUDA kernels developed for quantized calculations. For all training, we use 4x NVIDIA A100 80GB GPUs utilizing mixed precision (FP16) and gradient checkpointing to optimize memory usage.

The PPG (Photoplethysmography) encoder receives segments of input data corresponding to $T = 3000$ samples (24 seconds at $125Hz$) with 1 channel. There are 4 residual blocks arranged in order of $[64, 128, 256, 256]$ channels, each using a kernel size of $[15, 15, 15, 15]$. The cross-modal fusion module uses eight-heads of multi-head attention with an output dimension of 384.

The Chain-of-Thought Generator utilizes an adapted medical domain-specific tokenizer (32k vocabulary; BPE encoding) whose decoder weights were initialized using a pretrained biomedical language model (BioGPT [45]), followed by fine-tuning using clinician annotated reasoning traces. We thus term TrustCardio a "language-model-augmented multimodal framework" rather than a general-purpose multimodal LLM, as the reasoning decoder transfers a pretrained biomedical LM (BioGPT) rather than constructing a full multimodal LLM. Decoding uses a network of 6 layers of 512 hidden dimensions with 8 attention heads.

We optimize using AdamW with learning rates of $\eta_1 = 10^{-4}$ for Stage 1, $\eta_2 = 5 \times 10^{-5}$ for Stage 2, and $\eta_3 = 10^{-5}$

[2]https://physionet.org/content/mimic3wdb-matched/1.0/
[3]https://zenodo.org/record/7568783

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for Stage 3. The number of training epochs is $E_1 = 100$, $E_2 = 50$, and $E_3 = 30$. Validation performance is monitored to implement early stopping.

The target deployment of our edge-based model is as an edge-based product on an ARM Cortex-M7 (STMicroelectronics STM32H7 series), which includes 1MB Flash and 512KB SRAM, one of the most common modern smartwatch processors. Only the compressed prediction-and-uncertainty path (0.9 MB, Table VII) runs on the microcontroller; the transformer-based CoT decoder, exceeding Cortex-M7 limits, runs asynchronously on a paired smartphone or edge gateway, so the 31 ms latency refers to on-device vascular-age inference. Additionally, we test our model against the more powerful NVIDIA Jetson Nano to evaluate the difference between edge platforms.

E. Statistical Analysis

The Shapiro-Wilk test was initially used to confirm that the residuals were normally distributed. The two-tailed paired Student's t-test with Bonferroni correction was used when data were normally distributed($p > 0.05$). The Wilcoxon signed-rank test was a non-parametric alternative for situations in which data were not normally distributed. The primary family of 7 baselines $\times$ 4 metrics per dataset, subgroup analyses (Table IV), ablation studies (Table III), and outcome associations (Table XI) were all treated as specified exploratory analyses for analyses using unadjusted p-values for transparency. Effect sizes were calculated as Cohen's d. Confidence intervals (95%) were generated by bootstrapping, with 10,000 bootstrap resampling trials. Any p value less than 0.05, following corrections, represented a statistically significant result.

V. Experimental Results

A. Vascular Age Prediction Performance

The results presented in Table II show how TrustCardio compares to baseline algorithms on all three datasets used for benchmarking.

On the UK Biobank dataset, TrustCardio achieved MAE of 6.54 years, representing 9.0% improvement over CardioNet (MAE=7.19 years) with significantly higher correlation ($\rho = 0.57$ vs. 0.50). All improvements reached statistical significance ($p < 0.001$, paired t-test with Bonferroni correction; Cohen's $d = 0.31$ for the MAE reduction vs. CardioNet). Per-subgroup variability (mean $\pm$ 95% CI) appears in Table IV, and non-overlapping bootstrapped CIs for all primary comparisons are noted in Table II. Using bootstrapped 95% CIs (10,000 resamples), all TrustCardio to best baseline comparisons exhibited unique MAE intervals within all three data sets, as shown by representative per-subgroup CIs listed in Table IV. Across different population groups and data acquisition methods, improvements in performance were consistently observed in the MIMIC-III and PulseDB datasets through multimodal fusion based on personalized patient data. The R^2 values (0.24–0.32) are similar to the results from other studies that measured vascular age using PPG, which also found $R^2 \leq 0.30$ [2], [4]. Furthermore, the maximum R^2 reflects the physiological noise floor for single-PPG measurements, indicating that the strength of the clinical application of these models is more for population risk assessment than for predicting vascular age on an individual basis.

B. Ablation Studies

Table III presents ablation experiments quantifying the contribution of each TrustCardio component to overall performance.

Each component contributes incrementally: demographic features reduce MAE by 2.2%, cross-modal fusion adds 1.7%, and personalization contributes 1.6%. Uncertainty estimation enables confidence-aware predictions. The data augmentation pipeline further reduces MAE from 6.89 to 6.54 years (5.1%), representing the single largest individual contribution, primarily through improved robustness to signal degradation and more balanced subgroup learning.

C. Subgroup Analysis

Table IV breaks down the performance of the model across many different demographic and clinical sub-groups in order to assess the fairness of the model and to identify potential disparities.

D. Explainability Evaluation

1) Clinician Validation Study: A group of eight board-certified cardiologists reviewed a randomly selected set of 500 TrustCardio explanations. Each cardiologist used 5-point Likert scales to rate how clinically accurate, coherent, and actionable each explanation was. Those explanations that had a score of 4 or higher in all three categories were considered clinically appropriate. The results are shown in Table V.

TrustCardio has a Clinician Agreement Rate of 91.3% with substantial inter-rater reliability (Fleiss' κ=0.74) and an automated Reasoning Coherence Score of 0.87.

2) Comparison with Baseline Explainability Methods: Table VI compares TrustCardio's chain-of-thought explanation technique alongside traditional post-hoc explainability approaches that were applied to baseline model output.

The Chain-of-Thought strategy we use in TrustCardio provides much better performance than any other Post-Hoc explainer (for all Explainability Measurements). Furthermore, the fact that the CAR advantage (+23.2%) matches other research which found that generative explanations generated by models will convey superior clinical acceptability compared to post-hoc attributions [3], [26].

E. Edge Deployment Evaluation

1) Computational Efficiency: Table VII provides information regarding the computational efficiencies of TrustCardio variants designed to optimise performance based on their specific deployment scenario.

There are 4.2 million parameters in the TrustCardio model; the weight is equal to 16.8 MB and will take 892 million floating point operations (FLOPs) and 156 ms latency on the target microcontroller. The process of improving the model

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TABLE II
VASCULAR AGE PREDICTION PERFORMANCE COMPARISON ACROSS BENCHMARK DATASETS

Method	UK Biobank ($N = 98,672$)				MIMIC-III ($N = 10,282$)				PulseDB ($N = 5,361$)			
	MAE↓	RMSE↓	ρ↑	R^2↑	MAE↓	RMSE↓	ρ↑	R^2↑	MAE↓	RMSE↓	ρ↑	R^2↑
XGBoost-PPG [8]	8.42	10.51	0.41	0.16	9.87	12.34	0.32	0.09	8.91	11.23	0.38	0.13
LSTM-Attention [18]	7.98	9.87	0.44	0.19	9.21	11.56	0.36	0.12	8.34	10.47	0.41	0.16
ResNet1D-18 [17]	7.76	9.62	0.46	0.21	8.94	11.21	0.38	0.14	8.11	10.18	0.43	0.18
Transformer-PPG [30]	7.54	9.38	0.47	0.22	8.67	10.89	0.39	0.15	7.89	9.92	0.44	0.19
PPG-ViT [19]	7.41	9.21	0.48	0.23	8.52	10.71	0.40	0.16	7.74	9.73	0.45	0.20
Net1D [2]	7.27	9.04	0.49	0.24	8.38	10.54	0.41	0.17	7.61	9.56	0.46	0.21
CardioNet [20]	7.19	8.93	0.50	0.25	8.27	10.41	0.42	0.17	7.52	9.44	0.47	0.22
TrustCardio (Ours)	**6.54**	**8.12**	**0.57**	**0.32**	**7.58**	**9.53**	**0.49**	**0.24**	**6.87**	**8.63**	**0.54**	**0.29**
Improvement vs. best baseline	*9.0%*	*9.1%*	*14.0%*	*28.0%*	*8.3%*	*8.5%*	*16.7%*	*41.2%*	*8.6%*	*8.6%*	*14.9%*	*31.8%*

MAE: mean absolute error (years); RMSE: root mean square error (years); ρ: Pearson correlation; R^2: coefficient of determination.
All improvements are statistically significant ($p < 0.001$) based on paired t-tests with Bonferroni correction.
Bootstrapped 95% CIs (10,000 resamples) confirmed non-overlapping intervals for all primary comparisons (TrustCardio vs. CardioNet).

TABLE III
ABLATION STUDY ON UK BIOBANK DATASET

Configuration	MAE↓	RMSE↓	ρ↑	R^2↑
PPG Encoder Only	7.34	9.12	0.48	0.23
+ Demographic Features	7.18	8.91	0.50	0.25
+ Cross-Modal Fusion	7.06	8.76	0.51	0.26
+ Personalization Module	6.95	8.62	0.52	0.27
+ Uncertainty Estimation	6.89	8.54	0.53	0.28
+ Data Augmentation Pipeline	6.54	8.12	0.57	0.32
Full TrustCardio	**6.54**	**8.12**	**0.57**	**0.32**

Each row adds the specified component to the configuration above.

TABLE IV
SUBGROUP ANALYSIS ON UK BIOBANK DATASET

Subgroup	Category	MAE (years)	ρ
Sex	Male	6.58 ± 0.08	0.56
	Female	6.50 ± 0.07	0.58
Age Group	40–49 years	6.21 ± 0.11	0.52
	50–59 years	6.43 ± 0.09	0.55
	60–69 years	6.74 ± 0.10	0.58
	≥70 years	7.02 ± 0.14	0.60
BMI Category	Normal	6.37 ± 0.09	0.58
	Overweight	6.53 ± 0.08	0.57
	Obese	6.78 ± 0.10	0.55
Hypertension	No	6.34 ± 0.07	0.59
	Yes	6.86 ± 0.09	0.54
Diabetes	No	6.47 ± 0.07	0.58
	Yes	6.98 ± 0.13	0.52

Values reported as mean ± 95% confidence interval.

TABLE V
EXPLAINABILITY EVALUATION RESULTS

Metric	Value
Clinician Agreement Rate (CAR)	91.3%
Inter-rater Reliability (Fleiss' κ)	0.74
Clinical Accuracy Score	4.21 ± 0.68
Reasoning Coherence Score	4.08 ± 0.73
Actionability Score	3.94 ± 0.81
Reasoning Coherence Score (automated)	0.87
Feature Attribution Accuracy	84.6%

Scores reported as mean ± standard deviation on 5-point scale.
Inter-rater reliability indicates substantial agreement.

TABLE VI
EXPLAINABILITY METHOD COMPARISON

Method	CAR	RCS	FAA
SHAP (Net1D)	62.4%	0.58	71.2%
GradCAM (ResNet1D)	54.8%	0.52	65.7%
Attention Weights	68.1%	0.64	74.8%
LIME (XGBoost)	59.3%	0.55	68.4%
TrustCardio CoT	**91.3%**	**0.87**	**84.6%**

CAR: Clinician Agreement Rate; RCS: Reasoning Coherence Score; FAA: Feature Attribution Accuracy.

with quantization, distillation, and architecture search leads to the following results: the memory footprint of the updated model is 0.8 million parameters (0.9 MB) versus the original 4.2 million; and the FLOPs and latency reduction is 18.7× smaller (56 million FLOPs) and 5.0× faster (31 ms). This represents a total reduction in parameter count of 5.25×, storage compression of 18.7×, FLOPs reduction of 15.9×, and latency improvement of 5.0×. These figures correspond to the on-device prediction path; explanation generation is offloaded as described in Section IV-D and is excluded from the microcontroller budget.

2) Accuracy-Efficiency Trade-off: The trade-offs between computational efficiency and predictive accuracy are discussed in Table VIII.

TrustCardio-Edge incurs only 2.1% MAE increase (6.68 vs. 6.54 years) while achieving fivefold latency reduction, demonstrating the feasibility of deploying accurate vascular age prediction on consumer wearable devices.

3) Power Consumption Analysis: Table IX shows the power measurements calculated during a continuous inference test.

On the STM32H7 platform, TrustCardio-Edge consumes 89 mW during active inference and 12 mW idle, averaging 14.4 mW at 1 Hz duty cycle. This enables approximately 70 hours of continuous operation on a standard 250mAh smartwatch battery.

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TABLE VII
EDGE DEPLOYMENT EFFICIENCY ANALYSIS

Model Variant	Params	Size	FLOPs	Latency
Full Model (FP32)	4.2M	16.8 MB	892M	156 ms
QAT INT8	4.2M	4.3 MB	223M	98 ms
Distilled + INT8	1.1M	1.2 MB	78M	42 ms
NAS + Distilled + INT8	0.8M	0.9 MB	56M	31 ms
TrustCardio-Edge	0.8M	0.9 MB	56M	31 ms

Latency measured on ARM Cortex-M7 (STM32H7) at 480 MHz. Size represents flash memory footprint including weights and code.

TABLE VIII
ACCURACY-EFFICIENCY TRADE-OFF ANALYSIS

Model Variant	MAE	ΔMAE	Latency
Full Model (FP32)	6.54	–	156 ms
QAT INT8	6.59	+0.8%	98 ms
Distilled + INT8	6.66	+1.8%	42 ms
NAS + Distilled + INT8	6.68	+2.1%	31 ms
TrustCardio-Edge	6.68	+2.1%	31 ms

ΔMAE: relative increase in MAE compared to full model.
Latency measured on ARM Cortex-M7 (STM32H7) at 480 MHz.

TABLE IX
POWER CONSUMPTION ANALYSIS

Platform	Idle	Inference	Avg (1 Hz)
STM32H7 (Cortex-M7)	12 mW	89 mW	14.4 mW
Nordic nRF5340 (Cortex-M33)	8 mW	67 mW	9.8 mW
NVIDIA Jetson Nano	2.1 W	4.8 W	2.2 W

Avg (1 Hz): average power for 1 inference per second duty cycle. Measurements include processor, memory, and peripheral power.

TABLE X
CROSS-DATASET GENERALIZATION (MAE IN YEARS)

TrainTest	UK Biobank	MIMIC-III	PulseDB
UK Biobank	6.54	8.78	8.01
MIMIC-III	8.32	7.58	7.87
PulseDB	8.07	8.54	6.87
Combined	6.59	7.69	6.94

Diagonal entries (italicized) represent within-dataset performance.
Combined: training on pooled data from all three datasets.

F. Cross-Dataset Generalization

In Table X, we assess the generalizability across datasets by training on one dataset and testing on others.

Performance degrades across datasets by 1.2 - 2.5 years Mean Absolute Error (MAE) compared with evaluating the same dataset. This difference can be explained by a domain shift between two population groups and the way data was acquired.

G. Clinical Outcome Association

In order to assess the clinical relevance of this work, we will also examine the relationship between vascular age gap (VAG; predicted vascular age) and adverse cardiovascular events using the MIMIC-III dataset.

Patients with a VAG (Ventilation-Related Age) greater than nine years have a high risk of death (HR=1.89, $p < 0.001$), high rates of ICU readmission (HR=1.54, $p < 0.001$) and high rates of cardiovascular events (HR=1.72, $p < 0.001$), compared with patients with VAGs within the normal ranges. Conversely, patients whose VAGs are lower than -9 years have protective factors to help mitigate their risks. As a methodological caveat, TrustCardio's training is based upon chronological age and as a result the VAG reflects, in part, the residuals from chronological predicted values. Therefore, the hazard ratios in Table XI have been adjusted by chronological age, sex, BMI, comorbidities and severity of admission in order to reduce but not eliminate residual confounding factors. The above results support that the TrustCardio predictive system provides meaningful clinical prognostic information.

H. Comparison with Commercial Systems

Table XII shows the comparisons of TrustCardio to other commercially available PPG-based cardiovascular assessment systems.

TrustCardio uniquely combines superior prediction accuracy with chain-of-thought explainability, personalization, and edge deployment—capabilities not jointly offered by any existing commercial or academic system.

I. Sensitivity Analysis

1) Signal Quality Impact: Table XIII illustrates the effect of SQI on PPG performance as measured by the PPG signal quality index (SQI).

2) Demographic Covariate Importance: The table in Table XIV lists the results of the permutation importance analysis used to estimate how much weight should be placed on certain demographic attributes when evaluating demographic predictors.

The age of the individual is the primary explanatory factor of variability in the responses; the second most important factor is gender, and the third is body mass index (BMI). TrustCardio has been evaluated to ensure that there is no trivial age effect through the removal of chronological age from demographic information when evaluating performance. The MAE increased from 6.54 to 7.78 years (Δ+18.9%), consistent with the permutation importance of 0.342 in Table XIV; this indicates that PPG morphology is a major contributor and is independent from age in assisting with predictive accuracy.

VI. DISCUSSION

A. Clinical Implications

TrustCardio builds a foundation of trust in the area of AI-supported cardiovascular health monitoring in the consumer space. By enhancing the accuracy of predicting an individual's vascular age with clinician-aligned explanations, it will aid in developing more widespread support for clinical implementation because it allows healthcare providers to effectively validate the logic supporting the AI through established cardiovascular risk assessment methods.

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TABLE XI
ASSOCIATION BETWEEN VASCULAR AGE GAP AND CLINICAL OUTCOMES

Outcome	HR (95% CI)	p-value
VAG $>$ 9 years vs. -9 to 9 years (reference)		
In-hospital mortality	1.89 (1.62–2.21)	$<$0.001
ICU readmission	1.54 (1.31–1.82)	$<$0.001
Cardiovascular events	1.72 (1.48–2.01)	$<$0.001
VAG <-9 years vs. -9 to 9 years (reference)		
In-hospital mortality	0.71 (0.58–0.87)	0.001
ICU readmission	0.82 (0.68–0.98)	0.031
Cardiovascular events	0.76 (0.63–0.91)	0.003

HR: hazard ratio from Cox proportional hazards regression.
Adjusted for age, sex, BMI, comorbidities, and admission severity.

TABLE XII
COMPARISON WITH COMMERCIAL SYSTEMS

System	MAE	Explain.	Edge	Personal.
Apple Watch (vascular age)	9.2†	No	Yes	No
Fitbit (pulse wave)	8.8†	No	Yes	No
Samsung Health	8.5†	Partial	Yes	No
Academic SOTA [2]	7.57	No	No	No
TrustCardio	**6.54**	**Yes**	**Yes**	**Yes**

†Estimated from published validation studies.
Explain.: chain-of-thought explainability; Edge: on-device inference; Personal.: personalized modeling.

TABLE XIII
PERFORMANCE BY SIGNAL QUALITY LEVEL

SQI Range	N	MAE	ρ	Uncertainty
Excellent ($>$0.9)	41,234	6.52	0.57	4.21
Good (0.7–0.9)	38,921	6.87	0.53	5.34
Moderate (0.5–0.7)	14,872	7.34	0.48	6.78
Poor ($<$0.5)	3,645	8.21	0.41	8.92

SQI: signal quality index computed from morphological features.
Uncertainty: mean predicted standard deviation (years).

TABLE XIV
DEMOGRAPHIC FEATURE IMPORTANCE

Feature	Importance	ΔMAE (years)
Chronological Age	0.342	+1.24
Sex	0.156	+0.57
BMI	0.134	+0.49
Systolic BP	0.121	+0.44
Smoking Status	0.098	+0.36
Height	0.087	+0.32
Weight	0.062	+0.23

Importance: normalized permutation importance score.
ΔMAE: increase in MAE when feature is permuted.

B. Ethical Considerations

All datasets employed throughout this research remain available to the public with applicable ethical approvals and agreements for their usage within research purposes. The UK Biobank dataset was approved through the National Health Service (NHS), by way of the National Research Ethics Service (NRES-Ref:11/NW/0382). The MIMIC-III dataset has been published by PhysioNet through the Credentialed Health Data License agreement. PulseDB has been published under Creative Commons License at a 4.0 level.

TrustCardio is designed to assist rather than replace clinical decision-making. Uncertainty estimates are provided to reduce the risk of over-reliance on AI predictions.

Reproducibility Statement

Complete architecture specifications and hyperparameters are provided in Appendices A and B; all experiments use publicly available datasets with preprocessing and evaluation protocols fully described in Section IV.

VII. CONCLUSION

This research introduced the TrustCardio framework, an all-encompassing solution that combines large language model capabilities with multimodality and chain of thought explainability to enable edge-optimised deployment of personalised vascular age predication within consumer electronics used in the healthcare industry. TrustCardio utilises the systematic alignment of multimodal cross-modal fusion mechanisms, clinician-aligned explanation generation, and model compression techniques to create a new standard of prediction accuracy (Mean Absolute Error (MAE) of 6.54 years, a 9.0% improvement), highly explainable predictions (Clinician Agreement of 91.3%), and practical edge deployments for consumer electronics (ARM Cortex-M7, 31 ms inference latency).

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